

An Extended Elementary Proof of the Universal Approximation Theorem for Multilayer Perceptrons

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November 2025

Abstract

In theory, the Universal Approximation Theorem establishes that a Multilayer Perceptron (MLP) with a single hidden layer and a finite number of neurons can approximate any continuous function to any desired precision. This approximation capability is enabled by suitable activation functions such as Sigmoid, ReLU and Tanh. Building on the elementary proof framework introduced by Shaana A. and Karuna P. (BeyondAI 2024), this work extends their results by analysing limits rigorously, providing new visual explanations, and establishing a complete proof for ReLU-based piecewise simplex function constructions. Furthermore, we generalize the approach to the tanh activation function, laying groundwork towards a unified universal approximation framework for a broad class of activations. Computational validations and visualizations complement the theoretical results to enhance intuition and practical understanding. A companion article details the experimental investigation examining neuron efficiency and convergence rates in networks with single hidden layers.

Keywords: approximation, activation, continuity, convergence, ReLU, sigmoid, tanh.

1 Introduction

The Universal Approximation Theorem (UAT) provides a foundational guarantee that Multilayer Perceptrons (MLPs) with non-linear activations, most notably sigmoid or ReLU, can approximate any continuous function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Despite the significance of this theory in understanding the practical application of artificial neural networks (ANNs), existing proofs often involve complex mathematics that others may find difficult to fully comprehend and can obscure their intuition.

This paper builds upon the recent elementary proof framework by Shaana A. and Karuna P. [1], which provides accessible step-by-step constructions based on visual proof concepts from Michael Nielsen's proof approach [2]. Their work established approximation results for MLPs using sigmoid and ReLU activations and introduced piecewise function constructions utilizing simplices in the ReLU case.

Our contributions are:

1. A detailed rigorous analysis of the limits and approximation errors in the ReLU proof, closing gaps in the original argument.

2. An extension of the elementary proof to the tanh activation function, leveraging its relationship to sigmoid and adapting key constructions accordingly.
3. Visual and computational validations that bridge theoretical insights with empirical phenomena.
4. A path toward a generalized framework encompassing multiple activations, advancing the understanding of universal approximation in neural networks.

We begin by recalling the sigmoid case before presenting the tanh extension, followed by an in-depth treatment of ReLU. Throughout, we emphasize intuitive explanations supported by rigorous proofs and plain-language text.

2 Sigmoid Activation Proof

This section presents the proof of the Universal Approximation Theorem for Multilayer Perceptrons (MLPs) with sigmoid activation functions, following the elementary constructive framework of Shaana A. and Karuna P. [1]. The proof builds upon Nielsen’s visual explanation and rigorously demonstrates how sigmoid MLPs can approximate any continuous function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. It is divided into three core parts: step function approximation, construction of bump and tower functions, and approximation of continuous functions through piecewise constants.

2.1 Part 1: Approximating the Step Function via Sigmoid

The core idea is that the sigmoid function

$$\sigma(x) := \frac{1}{1 + e^{-x}}$$

can be used to approximate the step function $s(x)$, which is 0 for $x < 0$ and 1 for $x > 0$.

To approximate $s(x)$ by a scaled sigmoid function $\sigma(wx)$, it is essential to prove the pointwise convergence as the parameter $w \rightarrow \infty$.

2.1.1 Convergence for $x > 0$

We want to show that for every $\epsilon > 0$, there exists a threshold $W > 0$ such that for all $w > W$,

$$|\sigma(wx) - 1| < \epsilon.$$

Calculate the difference:

$$|\sigma(wx) - 1| = \left| \frac{1}{1 + e^{-wx}} - 1 \right| = \frac{1}{1 + e^{wx}}.$$

Define

$$A = e^x > 1,$$

so

$$|\sigma(wx) - 1| = \frac{1}{1 + A^w}.$$

To ensure this is less than ϵ , observe that

$$\frac{1}{1 + A^w} < \epsilon \iff 1 + A^w > \frac{1}{\epsilon} \iff A^w > \frac{1 - \epsilon}{\epsilon}.$$

Taking natural logarithms,

$$w > \frac{\ln \frac{1-\epsilon}{\epsilon}}{\ln A} =: W.$$

Therefore, for all $w > W$, the scaled sigmoid function is within ϵ of the step value 1.

2.1.2 Convergence for $x < 0$

Similar logic applies for $x < 0$, where we want to show that for every $\epsilon > 0$, there exists a threshold $W > 0$ such that for all $w > W$,

$$|\sigma(wx) - 0| = \sigma(wx) < \epsilon.$$

Calculate:

$$\sigma(wx) = \frac{1}{1 + e^{-wx}}.$$

For $x < 0$, define

$$A := e^{-x} > 1,$$

so

$$\sigma(wx) = \frac{1}{1 + A^w}.$$

To guarantee $\sigma(wx) < \epsilon$, we require

$$\frac{1}{1 + A^w} < \epsilon \implies A^w > \frac{1 - \epsilon}{\epsilon}.$$

Taking natural logarithms, this gives the threshold

$$w > \frac{\ln \frac{1-\epsilon}{\epsilon}}{\ln A} =: W.$$

Therefore, for all $w > W$, the scaled sigmoid is within ϵ of the step function value 0.

2.1.3 Extending a step function to n dimensions with a shifted step:

Incorporating a bias b , the step function's position/location can be shifted:

$$\sigma\left(w\left(x - \frac{b}{w}\right)\right) \approx s\left(x - \frac{b}{w}\right),$$

and this result extends to multiple n dimensions by setting the weights on all, but one input coordinate that defines the step direction, to zero, thereby approximating step functions along any coordinate direction.

2.2 Part 2: Constructing Bump and Tower Functions from Step Approximations

2.2.1 Bump functions:

We define bump functions as non-zero only on a finite interval (a, b) as follows:

$$c(x) := w_1 s_a(x) + w_2 s_b(x),$$

where s_a, s_b are step functions with transitions at points a and b respectively, and weights satisfy

$$w_1 = -w_2 = h > 0.$$

This creates a bump function of height h confined to the interval (a, b) :

$$c(x) = \begin{cases} 0, & x < a, \\ h, & a < x < b, \\ 0, & x > b. \end{cases}$$

2.2.2 Tower functions:

Extending bump functions into higher dimensions, tower functions combine bumps across multiple input coordinates:

$$t(\mathbf{x}) = \sum_j h_j \prod_{i=1}^n c_{i,j}(x_i),$$

where $c_{i,j}$ are bump functions along each coordinate axis and h_j are heights.

Applying the sigmoid with appropriate biases,

$$T(\mathbf{x}) = \sigma(t(\mathbf{x}) + b),$$

for large h and by selecting a bias b such that $-kh < b < -(k-1)h$, the function $t(x)$ is guaranteed to approximate a localized constant function of height kh , within the subset D_j of the input space. By creating highly localized approximations within specific subsets D_j of the input space, local constant-like regions (towers) are produced.

2.3 Part 3: Approximating Continuous Functions by Piecewise Constants

Given any continuous function

$$f : [0, 1]^n \rightarrow \mathbb{R}^m,$$

uniform continuity guarantees that for every $\epsilon > 0$, there exists a partition of the input domain into k^n equal cells $\{D_i\}$ such that for all $\mathbf{x}, \mathbf{y} \in D_i$,

$$\|f(\mathbf{x}) - f(\mathbf{y})\| < \epsilon.$$

Define the piecewise constant function

$$h(\mathbf{x}) = f(\mathbf{u}_i), \quad \mathbf{x} \in D_i,$$

where $\mathbf{u}_i$ are representative points in each cell.

Using the crucial Lemma A, for any $\mathbf{x} \in D_i$, the maximum distance to $\mathbf{u}_i$ is bounded by

$$\|\mathbf{x} - \mathbf{u}_i\| \leq \frac{\sqrt{n}}{k}.$$

By choosing k sufficiently large so that

$$\frac{\sqrt{n}}{k} < \delta,$$

where δ corresponds to the ϵ of continuity, the approximation error

$$\|f(\mathbf{x}) - h(\mathbf{x})\| \leq \epsilon$$

holds uniformly for all $\mathbf{x}$, proving h approximates f arbitrarily well.

This extends from scalar to vector-valued functions by approximating each component separately and stacking outputs.

This completes the rigorous, constructive proof of universal approximation for sigmoid-activated MLPs as developed by Shaana A. and Karuna P. [1].

3 Extension to Tanh Activation Proof

The tanh function, $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, is used to demonstrate universal approximation by showing it approximates a step function from -1 to 1. This extension then relies directly on the algebraic structure derived for the sigmoid function. Afterwards, the construction of bump and tower functions, and approximation of continuous functions through piecewise constants are carried out.

3.1 Approximating the Step Function via Tanh

The exponential form of the hyperbolic tangent function, defined as,

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}},$$

can be used to approximate the step function $s(x)$, which takes the value -1 for $x < 0$ and 1 for $x > 0$.

To approximate $s(x)$ by a scaled tanh function $\tanh(wx)$, it is essential to prove the pointwise convergence as the parameter $w \rightarrow \infty$.

3.1.1 Convergence for $x > 0$

We want to show that for every $\epsilon > 0$, there exists a threshold $W > 0$ such that for all $w > W$,

$$|\tanh(wx) - 1| < \epsilon.$$

Calculate the difference:

$$|\tanh(wx) - 1| = \left| \frac{e^{wx} - e^{-wx}}{e^{wx} + e^{-wx}} - 1 \right| = 1 - \frac{e^{wx} - e^{-wx}}{e^{wx} + e^{-wx}} = \frac{2e^{-wx}}{e^{wx} + e^{-wx}}.$$

We want to find W such that for all $w > W$,

$$\frac{2e^{-wx}}{e^{wx} + e^{-wx}} < \epsilon.$$

Rewrite as:

$$2e^{-wx} < \epsilon(e^{wx} + e^{-wx}) = \epsilon e^{wx} + \epsilon e^{-wx}.$$

Bring all terms to one side:

$$2e^{-wx} - \epsilon e^{wx} - \epsilon e^{-wx} < 0.$$

Group terms:

$$(2 - \epsilon)e^{-wx} - \epsilon e^{wx} < 0.$$

Divide both sides by the positive e^{-wx} :

$$2 - \epsilon < \epsilon e^{2wx}.$$

This implies:

$$e^{2wx} > \frac{2 - \epsilon}{\epsilon}.$$

Taking natural logarithms:

$$2wx > \ln \left(\frac{2 - \epsilon}{\epsilon} \right).$$

Finally,

$$w > \frac{1}{2x} \ln \left(\frac{2 - \epsilon}{\epsilon} \right).$$

Therefore, for all $w > W$, the scaled tanh function is within ϵ of the step value 1.

3.1.2 Convergence for $x < 0$

Similarly, we want to show that for every $\epsilon > 0$, there exists a threshold $W > 0$ such that for all $w > W$,

$$|\tanh(wx) + 1| < \epsilon.$$

Calculate the difference:

$$|\tanh(wx) + 1| = \left| \frac{e^{wx} - e^{-wx}}{e^{wx} + e^{-wx}} + 1 \right| = \frac{2e^{wx}}{e^{wx} + e^{-wx}}.$$

We want to find W such that for all $w > W$,

$$\frac{2e^{wx}}{e^{wx} + e^{-wx}} < \epsilon.$$

Rewrite the inequality as:

$$2e^{wx} < \epsilon(e^{wx} + e^{-wx}) = \epsilon e^{wx} + \epsilon e^{-wx}.$$

Bring all terms to one side:

$$2e^{wx} - \epsilon e^{wx} - \epsilon e^{-wx} < 0.$$

Group terms:

$$(2 - \epsilon)e^{wx} - \epsilon e^{-wx} < 0.$$

Divide both sides by the positive quantity e^{-wx} , since $x < 0$ implies $e^{-wx} > 0$:

$$(2 - \epsilon)e^{2wx} < \epsilon.$$

Rearranged as:

$$e^{2wx} < \frac{\epsilon}{2 - \epsilon}.$$

Taking natural logarithms gives:

$$2wx < \ln \left(\frac{\epsilon}{2 - \epsilon} \right).$$

Since $x < 0$, set $x = -|x|$, thus:

$$2w(-|x|) < \ln \left(\frac{\epsilon}{2 - \epsilon} \right) \implies -2w|x| < \ln \left(\frac{\epsilon}{2 - \epsilon} \right).$$

Multiply both sides by -1 , to reverse inequality:

$$2w|x| > -\ln \left(\frac{\epsilon}{2 - \epsilon} \right) = \ln \left(\frac{2 - \epsilon}{\epsilon} \right).$$

Finally, solve for w :

$$w > \frac{1}{2|x|} \ln \left(\frac{2 - \epsilon}{\epsilon} \right).$$

Therefore, for all $w > W$, with

$$W := \frac{1}{2|x|} \ln \left(\frac{2 - \epsilon}{\epsilon} \right),$$

the scaled $\tanh(wx)$ is within ϵ of the value -1 .

3.1.3 Extension to $\mathbb{R}^n$ via Tanh

Incorporating a bias b , the $\tanh$ step function's position or location can be shifted as follows:

$$\tanh \left(w \left(x - \frac{b}{w} \right) \right) \approx s \left(x - \frac{b}{w} \right),$$

where

$$s(x) := \begin{cases} -1, & x < 0, \\ 1, & x > 0, \end{cases}$$

is the ideal step function taking values in $\{-1, 1\}$.

This result extends to multiple n dimensions by introducing a bias parameter b and choosing weights to focus on a single input dimension, the step function approximation extends to any coordinate direction in $\mathbb{R}^n$.

3.2 Rewriting Tanh in a Sigmoid-Like Form

Recall that the hyperbolic tangent function admits the exponential form and is defined as:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}},$$

It satisfies the key relation to the sigmoid function $\sigma(x) = \frac{1}{1+e^{-x}}$, as

$$\begin{aligned} \tanh(x) &= \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1} \\ &= \frac{\frac{e^{2x}+1}{2} - \frac{2}{e^{2x}+1}}{\frac{e^{2x}+1}{2} + \frac{2}{e^{2x}+1}} \\ &= 2 \cdot \frac{e^{2x}}{e^{2x} + 1} - 1 \end{aligned}$$

$$\tanh(x) = 2\sigma(2x) - 1.$$

This relationship allows us to adapt the sigmoid-based elementary proof to $\tanh$ by suitable scaling and shifting of parameters.

3.3 Constructing Bump and Tower Functions from Step Approximations

3.3.1 Bump Functions Constructed from the tanh Activation

Consider real numbers $a < b$ and a scaling parameter $w > 0$. Define the bump function using tanh as

$$a_w(x) := \frac{1}{2} \left(\tanh(w(x - a)) - \tanh(w(x - b)) \right).$$

Using the algebraic identity

$$\tanh(x) = 2\sigma(2x) - 1,$$

rewrite $a_w(x)$:

$$\begin{aligned} a_w(x) &= \frac{1}{2} (2\sigma(2w(x - a)) - 1 - (2\sigma(2w(x - b)) - 1)) \\ &= \sigma(2w(x - a)) - \sigma(2w(x - b)). \end{aligned}$$

The known convergence for sigmoid bumps states that

$$\lim_{w \rightarrow \infty} \sigma(2w(x - a)) - \sigma(2w(x - b)) = \chi_{(a,b)}(x),$$

where $\chi_{(a,b)}$ is the indicator function of the interval (a, b) .

Hence, we obtain the limit for the tanh bump function:

$$\lim_{w \rightarrow \infty} a_w(x) = \chi_{(a,b)}(x) = \begin{cases} 1, & a < x < b, \\ 0, & \text{otherwise.} \end{cases}$$

For fixed x , the evaluation is:

$$a_w(x) = \begin{cases} \frac{1}{2}(-1 - (-1)) = 0, & x < a, \\ \frac{1}{2}(1 - (-1)) = 1, & a < x < b, \\ \frac{1}{2}(1 - 1) = 0, & x > b, \end{cases}$$

showing a_w acts as a “smooth step difference” bump localized on (a, b) .

3.3.2 Tower Functions Constructed from Multiple tanh Bumps

Consider an input vector $\mathbf{x} = (x_1, x_2, \dots, x_n)$, intervals (a_i, b_i) for each coordinate, and scaling parameters $w_i > 0$.

Define the one-dimensional bump functions as before:

$$a_{w_i,i}(x_i) := \frac{1}{2} \left(\tanh(w_i(x_i - a_i)) - \tanh(w_i(x_i - b_i)) \right).$$

Using the sigmoid identity,

$$a_{w_i,i}(x_i) = \sigma(2w_i(x_i - a_i)) - \sigma(2w_i(x_i - b_i)),$$

where σ is the sigmoid function.

Construct the multidimensional tower function as the product of bumps:

$$t_{\mathbf{w}}(\mathbf{x}) := \prod_{i=1}^n a_{w_i, i}(x_i) = \prod_{i=1}^n [\sigma(2w_i(x_i - a_i)) - \sigma(2w_i(x_i - b_i))].$$

By the limit behaviour of the sigmoid,

$$\lim_{w_i \rightarrow \infty} \sigma(2w_i(x_i - a_i)) - \sigma(2w_i(x_i - b_i)) = \chi_{(a_i, b_i)}(x_i),$$

the indicator function of (a_i, b_i) .

Hence, the tower function converges pointwise to the indicator function of the hyper rectangle:

$$\lim_{\min w_i \rightarrow \infty} t_{\mathbf{w}}(\mathbf{x}) = \prod_{i=1}^n \chi_{(a_i, b_i)}(x_i) = \chi_{\prod_{i=1}^n (a_i, b_i)}(\mathbf{x}).$$

This means $t_{\mathbf{w}}(\mathbf{x})$ is positive exactly within $\prod_i (a_i, b_i)$ and zero outside, with the steepness controlled by $\mathbf{w}$.

Thus, the tower functions constructed using tanh activations correspond algebraically to products of sigmoid-based bump functions, with output affine transformations implied by the tanh-sigmoid identity. This construction directly inherits the convergence properties of sigmoid bumps and extends to multidimensional piecewise constant approximations.

3.4 Approximating Continuous Functions by Piecewise Constants

Let $f : [0, 1]^n \rightarrow \mathbb{R}^m$ be continuous.

By uniform continuity, for any $\epsilon > 0$, there exists $\delta > 0$ such that for all $\mathbf{x}, \mathbf{y} \in [0, 1]^n$,

$$\|\mathbf{x} - \mathbf{y}\| < \delta \implies \|f(\mathbf{x}) - f(\mathbf{y})\| < \epsilon.$$

Partition the domain $[0, 1]^n$ into k^n hypercubes $\{D_i\}$ of side length $1/k < \delta/\sqrt{n}$, where

$$D_i = \left\{ \mathbf{x} : \frac{u_{i,j}}{k} \leq x_j < \frac{u_{i,j} + 1}{k}, \quad j = 1, \dots, n \right\}$$

for integer vectors $\mathbf{u}_i = (u_{i,1}, \dots, u_{i,n}) \in \{0, \dots, k-1\}^n$.

Define the piecewise constant function

$$h(\mathbf{x}) = f\left(\frac{\mathbf{u}_i}{k}\right), \quad \mathbf{x} \in D_i.$$

By construction, the approximation error satisfies

$$\|f(\mathbf{x}) - h(\mathbf{x})\| < \epsilon, \quad \forall \mathbf{x} \in [0, 1]^n.$$

Approximation of Indicator Functions χ_{D_i}

Using the identity

$$\tanh(x) = 2\sigma(2x) - 1,$$

each one-dimensional bump function along coordinate j is

$$a_{w_j,j}(x_j) = \frac{1}{2} (\tanh(w_j(x_j - a_j)) - \tanh(w_j(x_j - b_j))) = \sigma(2w_j(x_j - a_j)) - \sigma(2w_j(x_j - b_j)),$$

which pointwise converges to $\chi_{(a_j,b_j)}(x_j)$ as $w_j \rightarrow \infty$.

The multidimensional tower function approximating χ_{D_i} is the product of bumps,

$$t_{\mathbf{w}^i}(\mathbf{x}) = \prod_{j=1}^n a_{w_j^i,j}(x_j),$$

which converges pointwise to the indicator $\chi_{D_i}(\mathbf{x})$ as $\min_j w_j^i \rightarrow \infty$.

3.4.1 Full Approximation as Finite Sum of Towers

The piecewise function $h(\mathbf{x})$ can therefore be expressed as

$$h(\mathbf{x}) \approx \sum_i f\left(\frac{\mathbf{u}_i}{k}\right) t_{\mathbf{w}^i}(\mathbf{x}),$$

where each tower function $t_{\mathbf{w}^i}$ is realizable by a tanh-activated multilayer perceptron with appropriately large weights $\mathbf{w}^i$.

3.4.2 Extension to Vector-Valued Targets

For vector-valued $f : [0, 1]^n \rightarrow \mathbb{R}^m$, approximate each component individually:

$$h(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}), \dots, h_m(\mathbf{x})),$$

where each scalar h_j is constructed using the same procedure applied to the scalar component f_j .

4 Approximation by ReLU Activation

The ReLU activation function is defined as $\varphi(x) = \max(0, x)$. A key property enabling approximation is that ReLU networks inherently produce piecewise linear functions, which can approximate continuous functions on compact domains via piecewise linear functions.

4.1 Approximation of Continuous Functions $f : [0, 1] \rightarrow \mathbb{R}$

Following the constructive elementary framework of Shaana A. and Karuna P. [1], we present a rigorous proof that piecewise linear functions generated by ReLU MLPs can approximate any continuous function $f : [0, 1] \rightarrow \mathbb{R}$ arbitrarily well.

Let f be a continuous function on the interval $[0, 1]$. Partition this interval into n subintervals defined by points

$$0 = x_0 < x_1 < \cdots < x_n = 1,$$

where the mesh size $\max_i(x_{i+1} - x_i)$ approaches zero as $n \rightarrow \infty$.

On each subinterval $[x_i, x_{i+1}]$, define the affine interpolant

$$l_i(x) = f(x_i) + \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}(x - x_i).$$

Construct the piecewise linear function h_n by setting

$$h_n(x) = l_i(x) \text{ for } x \in [x_i, x_{i+1}],$$

which can also be expressed in the form

$$h_n(x) = \sum_{i=0}^n c_i \max(0, x - x_i),$$

with coefficients c_i chosen to satisfy interpolation conditions $h_n(x_i) = f(x_i)$.

Due to the uniform continuity of f , for every $\varepsilon > 0$, there exists N such that for all $n > N$,

$$\|f - h_n\|_\infty < \varepsilon.$$

This shows h_n converges uniformly to f . Moreover, each such piecewise linear function h_n can be exactly realized by a ReLU MLP with a single hidden layer.

The detailed decomposition of the approximation error, following Shaana A. and Karuna P., expresses the error on each subinterval $[x_j, x_{j+1}]$ as

$$f(x) - h_n(x) = (f(x) - f(x_j))(x_{j+1} - x)/(x_{j+1} - x_j) + (f(x) - f(x_{j+1}))(x - x_j)/(x_{j+1} - x_j),$$

which is bounded by the maximal deviation from the interpolation points and tends to zero as the partition refines.

Hence, this constructive argument establishes that ReLU MLPs approximate continuous functions on $[0, 1]$ via piecewise linear functions converging uniformly to the target.

4.2 Extension to Higher Dimensions via Simplices: Construction and Uniform Convergence

This proof that details how ReLU-activated MLPs can be constructed in higher dimensions via simplices closes the key assumption made in last year's work [1], rigorously establishing uniform approximation on compact domains.

Let $f : [0, 1]^n \rightarrow \mathbb{R}$ be continuous.

Partition the domain into simplices $\{S_j\}$, each given by vertices $\{v_0^{(j)}, \dots, v_n^{(j)}\}$.

Express any point $\mathbf{x} \in S_j$ as barycentric combination:

$$\mathbf{x} = \sum_{i=0}^n \alpha_i^{(j)}(\mathbf{x}) v_i^{(j)}, \quad \sum_i \alpha_i^{(j)}(\mathbf{x}) = 1, \quad \alpha_i^{(j)} \geq 0.$$

where the weights $\alpha_i^{(j)}(\mathbf{x})$ represent the relative contribution of each vertex $v_i^{(j)}$ to $\mathbf{x}$.

Define piecewise affine approximation:

$$g_j(\mathbf{x}) = \sum_{i=0}^n \alpha_i^{(j)}(\mathbf{x}) f(v_i^{(j)}), \quad \mathbf{x} \in S_j,$$

and set

$$g(\mathbf{x}) = g_j(\mathbf{x}) \quad \text{if } \mathbf{x} \in S_j,$$

which converges uniformly to f as the simplex mesh refines.

4.2.1 Representation of Simplices and Affine Functions with ReLU

Each simplex S_j is characterized by linear inequalities:

$$h_i^{(j)}(\mathbf{x}) := \mathbf{w}_i^{(j)} \cdot \mathbf{x} + b_i^{(j)} \geq 0, \quad i = 0, \dots, n,$$

defining the facet half-spaces.

Using ReLU activations:

$$r_i^{(j)}(\mathbf{x}) := \max\{0, h_i^{(j)}(\mathbf{x})\}$$

can be used to implement these half-space constraints.

Affine functions on S_j can be reconstructed via linear combinations of these ReLU activations:

$$g_j(\mathbf{x}) = \sum_{i=0}^n c_i^{(j)} r_i^{(j)}(\mathbf{x}),$$

where coefficients $c_i^{(j)}$ are chosen to match the function values $f(v_i^{(j)})$ at vertices.

The global piecewise function $g(\mathbf{x})$ is then realized by summing these affine-part-implementing ReLU compositions over all simplices.

4.2.2 Convergence and Approximation Error Bound

For any $\mathbf{x} \in S_j$, the pointwise approximation error is bounded by:

$$\begin{aligned} |f(\mathbf{x}) - g(\mathbf{x})| &= \left| f(\mathbf{x}) - \sum_{i=0}^n \alpha_i^{(j)}(\mathbf{x}) f(v_i^{(j)}) \right| \\ &= \left| \sum_{i=0}^n \alpha_i^{(j)}(\mathbf{x}) (f(\mathbf{x}) - f(v_i^{(j)})) \right|. \end{aligned}$$

By the triangle inequality and the fact that $\alpha_i^{(j)}(\mathbf{x}) \geq 0$ and $\sum_i \alpha_i^{(j)}(\mathbf{x}) = 1$, we have

$$|f(\mathbf{x}) - g(\mathbf{x})| \leq \sum_{i=0}^n \alpha_i^{(j)}(\mathbf{x}) |f(\mathbf{x}) - f(v_i^{(j)})| \leq \max_{0 \leq i \leq n} |f(\mathbf{x}) - f(v_i^{(j)})|.$$

By continuity of f , for every $\epsilon > 0$ there exists $\delta > 0$ such that if

$$\text{diameter}(S_j) := \sup_{\mathbf{x}, \mathbf{y} \in S_j} \|\mathbf{x} - \mathbf{y}\| < \delta,$$

then

$$|f(\mathbf{x}) - g(\mathbf{x})| < \epsilon \quad \text{for all } \mathbf{x} \in S_j.$$

Thus, uniform convergence follows as the simplicial mesh refines (diameter shrinks).

4.2.3 Extension to Vector-Valued Functions

For vector-valued $f : [0, 1]^n \rightarrow \mathbb{R}^m$, approximate each component f_k , $k = 1, \dots, m$, individually:

$$g_k(\mathbf{x}) = \sum_{i=0}^n \alpha_i^{(j)}(\mathbf{x}) f_k(v_i^{(j)}),$$

and define

$$g(\mathbf{x}) = (g_1(\mathbf{x}), \dots, g_m(\mathbf{x})),$$

which converges uniformly component-wise.

4.2.4 Conclusion

These constructive steps rigorously prove that ReLU MLPs with finite neurons and layers can approximate any continuous function uniformly on compact domains by piecewise linear functions on refined simplicial partitions. This completes the constructive proof in higher dimensions, closing the essential simplicial assumption mentioned by Shaana A. and Karuna P. [1].

References

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